20. The number of ways to arrange the letters of the word " CHEESE "

(a) 120 (b) 240 (c) 720 (d) 6

21. Thirteen guests has participated in a dinner. The number of handshakes happened in the dinner is

(a) 715 (b) 78

(c) 286 (d) 13

22. Number of words with or without meaning that can be formed using letters of the word "EQUATION" , with no repetition of letters is

(a) 7! (b) 3! (c) 8! (d) 5!

23. Sum of Binomial co-efficient in a particular expansion is 256, then number of terms in the expansion is

(a) 8 (b) 7

(c) 6 (d) 9

24. The number of permutation of n different things taken r at a time, when the repetition is allowed is

(a) r^n (b) n^r

(c) $\frac{n!}{(n-r)!}$ (d) $\frac{n!}{(n+r)!}$

25. Sum of the binomial coefficients is

(a) 2^n (b) n^2

(c) $2n$ (d) n+17

Miscellaneous Problems

1. Resolve into Partial Fractions : $\frac{5x+7}{(x-1)(x+3)}$

2. Resolve into Partial Fractions: $\frac{x-4}{x^2-3x+2}$

3. Decompose into Partial Fractions: $\frac{6x^2-14x-27}{(x+2)(x-3)^2}$

4. Decompose into Partial Fractions: $\frac{5x^2-8x+5}{(x-2)(x^2-x+1)}$

5. Evaluate the following.

(i) $\frac{3! \times 0! + 0!}{2!}$ (ii) $\frac{3! + 1!}{(2^2)!}$

(iii) $\frac{(3!)! \times 2!}{5!}$

6. How many code symbols can be formed using 5 out of 6 letters A, B, C, D, E, F so that the letters a) cannot be repeated b) can be repeated c) cannot be repeated but must begin with E d) cannot be repeated but end with CAB.

7. From 20 raffle tickets in a hat, four tickets are to be selected in order. The holder of the first ticket wins a car, the second a motor cycle, the third a bicycle and the fourth a skateboard. In how many different ways can these prizes be awarded?

8. In how many different ways, 2 Mathematics, 2 Economics and 2 History books can be selected from 9 Mathematics, 8 Economics and 7 History books?

9. Let there be 3 red, 2 yellow and 2 green signal flags. How many different signals are possible if we wish to make signals by arranging all of them vertically on a staff?

10. Find the Co-efficient of x^{11} in the expansion of $\left(x+\frac{2}{x^2}\right)^{17}$

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